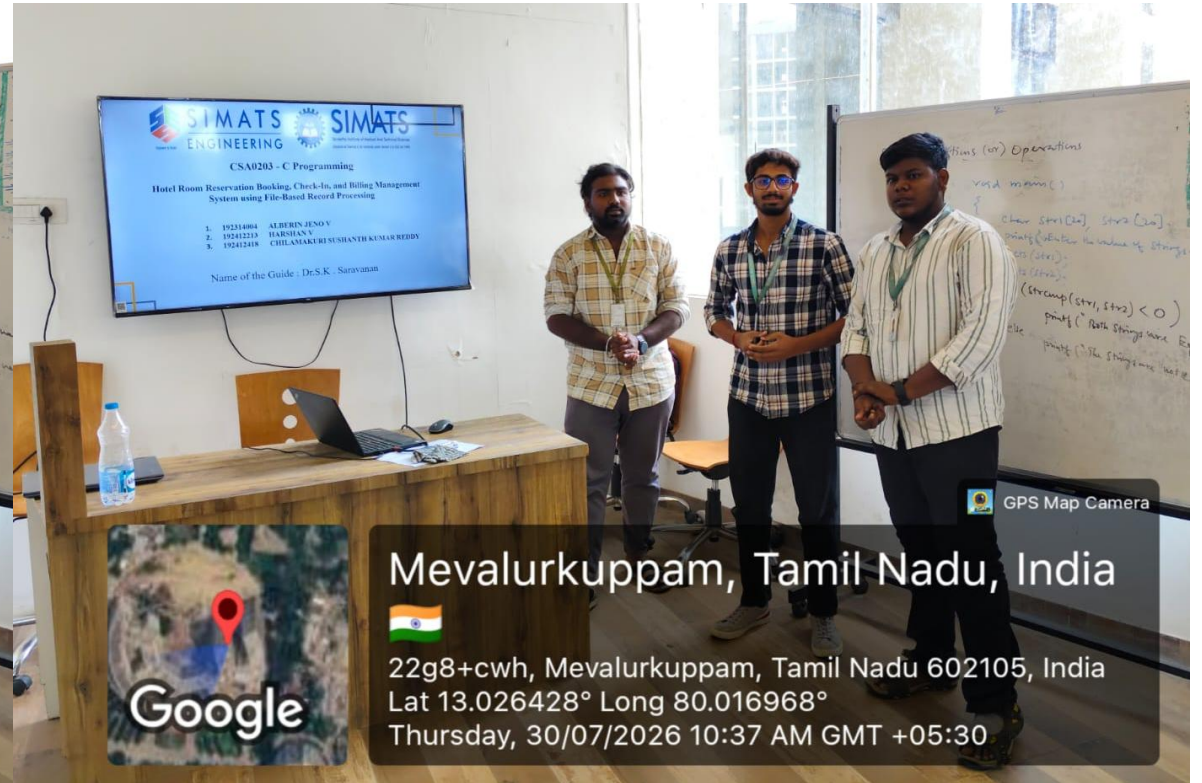
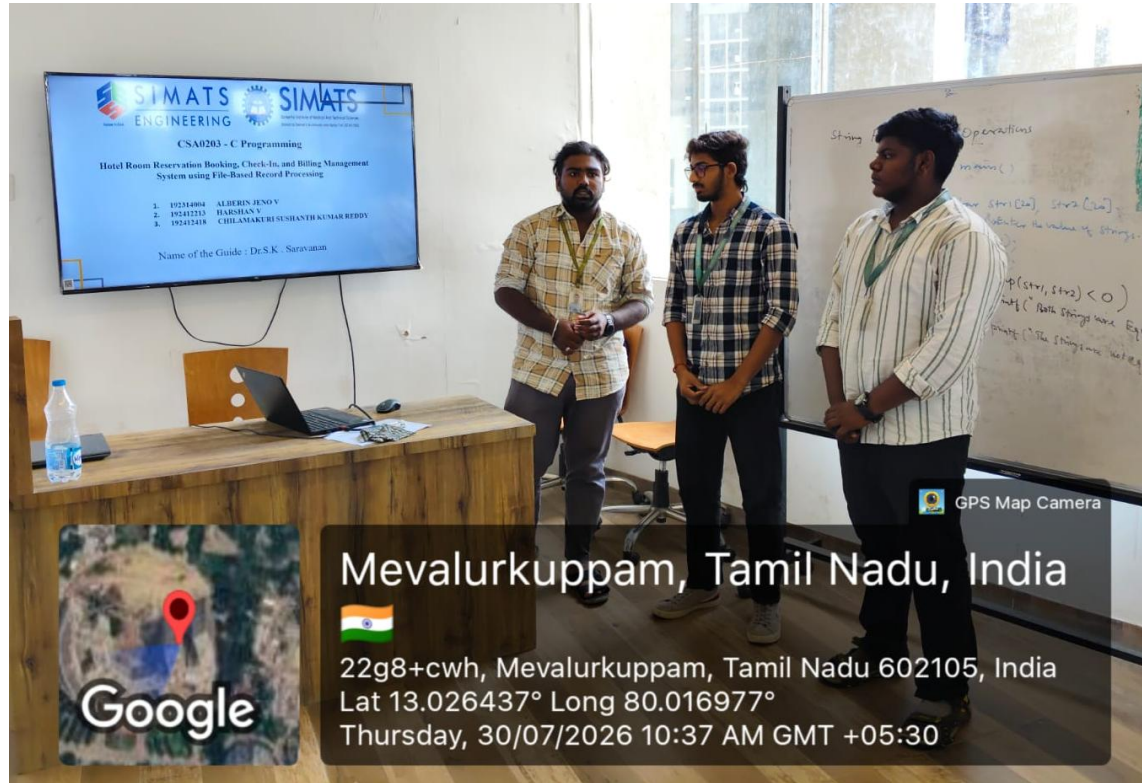






SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0203 - C Programming

Hotel Room Reservation Booking, Check-In, and Billing Management System using File-Based Record Processing

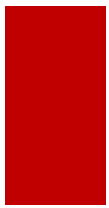
1. 192314004 ALBERIN JENO V
2. 192412213 HARSHAN V
3. 192412418 CHILAMAKURI SUSHANTH KUMAR REDDY

Name of the Guide : Dr.S.K . Saravanan



INTRODUCTION

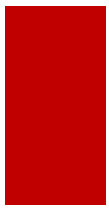
- ✓ Hotels require an efficient system to manage room reservations, guest check-in/check-out, and billing.
- ✓ Traditional manual methods are time-consuming, prone to errors, and difficult to maintain.
- ✓ This project is developed using the **C programming language** with **file handling** to store guest and booking records permanently.
- ✓ The system automates room booking, availability tracking, guest management, and bill generation through a menu-driven interface.
- ✓ It improves accuracy, reduces manual effort, and demonstrates the practical use of **structures, functions, and file-based record processing** in C programming.



PROBLEM STATEMENT



- ✓ Manual hotel management is slow and error-prone.
- ✓ Room booking and availability tracking are difficult.
- ✓ Guest records are hard to store and retrieve.
- ✓ Billing calculations may contain errors.
- ✓ A file-based C application is needed to automate reservations, check-in/check-out, and billing while maintaining permanent records.



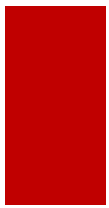
PROPOSED SOLUTION



The proposed system is a menu-driven application developed in C programming that automates hotel operations using file-based record processing. It manages room reservations, guest check-in/check-out, and billing while securely storing records for future access.

Key Features

- Room reservation and availability management
- Guest check-in and check-out
- Automatic billing and payment calculation
- File-based storage for permanent records
- Search, update, and delete guest information
- User-friendly menu-driven interface

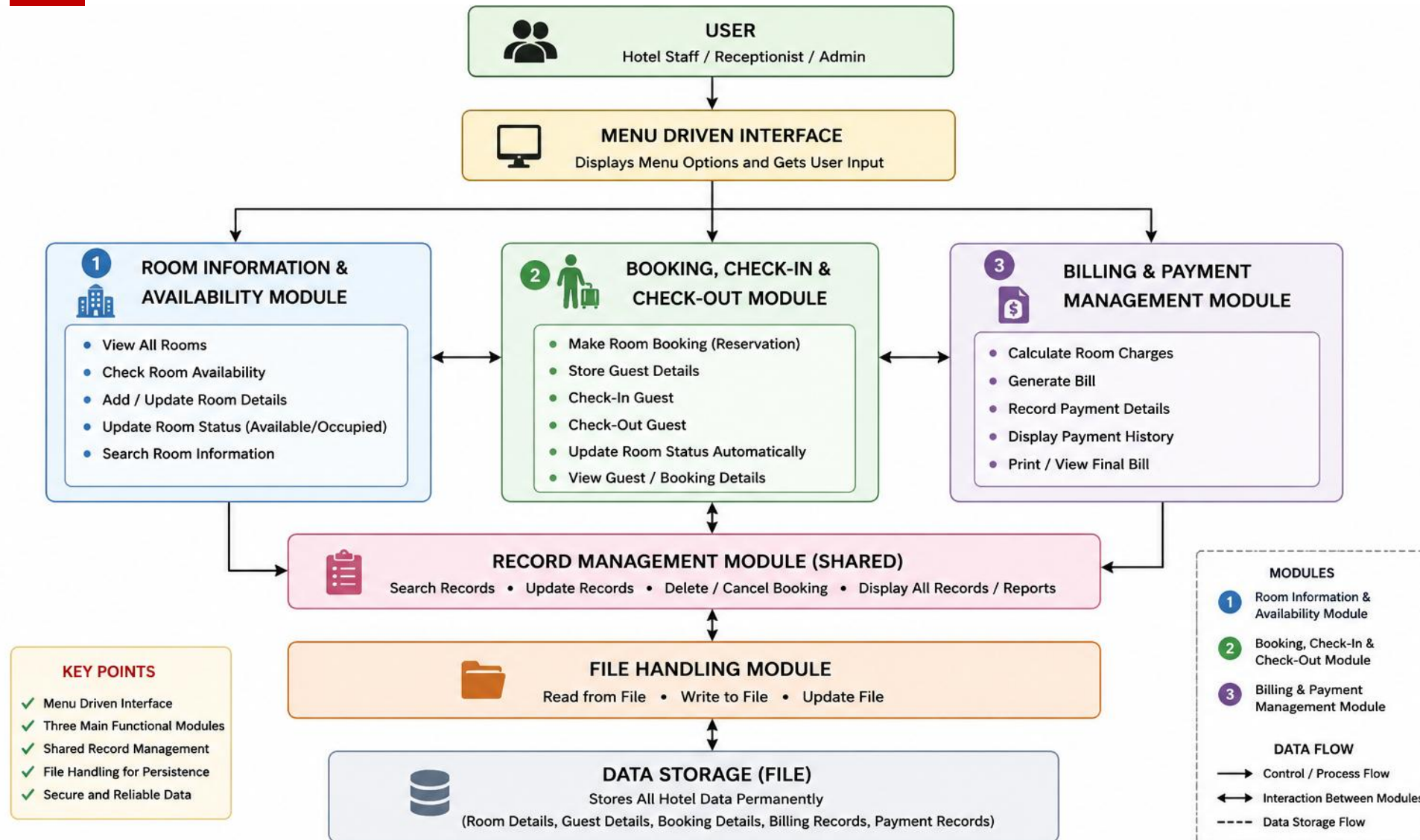


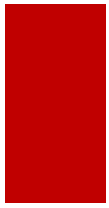
Functionalities



- **Room Information & Availability** View available and occupied rooms.
- Update room status after booking or check-out.
- **Booking Management** Reserve rooms for guests.
- Store guest details using file handling.
- **Check-In & Check-Out** Register guest check-in.
- Process check-out and update room availability.
- **Billing & Payment** Calculate room charges automatically.
- Generate and display the final bill.
- **Record Management** Search, update, delete, and display guest records.
- **File-Based Storage** Save and retrieve records permanently using files.

SYSTEM ARCHITECTURE

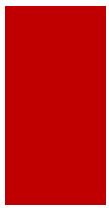




CONCLUSION



The Hotel Room Reservation Booking, Check-In, and Billing Management System demonstrates how file-based record processing can be used to automate essential hotel operations using the C programming language. The system efficiently manages room reservations, guest check-in and check-out, billing, and record maintenance through a simple menu-driven interface. By utilizing structures, functions, and file handling, it ensures accurate data management, reduces manual errors, and provides permanent storage of records. This project serves as a practical application of fundamental C programming concepts and can be further enhanced with features such as user authentication, database integration, and an online reservation interface.



REFERENCES

Balagurusamy, E. (2019). *Programming in ANSI C* (8th ed.). McGraw Hill Education.

Gottfried, B. S. (2018). *Programming with C* (4th ed.). McGraw Hill Education.

Kanetkar, Y. (2020). *Let Us C* (17th ed.). BPB Publications.

Kernighan, B. W., & Ritchie, D. M. (1988). *The C programming language* (2nd ed.). Prentice Hall.

Schildt, H. (2000). *C: The complete reference* (4th ed.). McGraw-Hill.

International Organization for Standardization. (2018). *Programming languages—C (ISO/IEC 9899:2018)*.

THANK YOU

EVERYONE FOR YOUR PATIENCE